

GiuliaD'Angelo

Contacts

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Languages

Italian (mother tongue)
English (proficiency)

Current Position

Marie Skłodowska-Curie
Postdoctoral Fellow -
Department of
Cybernetics, Faculty of
Electrical Engineering,
Czech Technical
University in Prague

Software skills

Programming Languages

Python, C++, Matlab,
JavaScript, HTML, Brian,
Neuron

Middleware & Tools

Yarp, ROS

Neuromorphic platforms

ATIS, SpiNNaker, Speck,
Loihi

Tools & Libraries

PyTorch, snnTorch,
sinabs, Tonic,
Metavision, bimvee

Personal profiles

[Google scholar](#)

[ORCID](#)

SCOPUS ID:

57215559435

[Linkedin](#)

[Twitter](#)

[Github](#)

Personal Interest

I focus on
high-performance active
visual algorithms for
online robotic
applications. I
developed attention
mechanisms using
intensity, motion
estimation, and depth
cues to enable robotic
interaction through
saccadic eye
movements, leveraging
neuromorphic sensing
and computing.

I have nine years of **research and development** experience in low-power, low-latency **computer vision** applications for real-time robotics. I have designed, implemented, and led a wide range of projects leveraging spiking neural networks in combination with state-of-the-art bioinspired event-based cameras and neuromorphic computing for **embodied active vision systems**.

Work Experience

2024 - NOW	Marie Skłodowska-Curie Postdoctoral Fellow Czech Technical University (Prague, Czechia) Embodied Active Vision - EveNt DrivEn Active Vision for Object peRception (EN-DEAVOR) - Event-based vision sensing and neuromorphic computing. Principal Investigator & Project leader.
2022 - 2024	Postdoctoral Researcher - Computer Vision for Robotics Italian Institute of Technology (Italy) Event Driven Perception for Robotics - R&D & Project management
May-June, 2023	Visiting Researcher - Embodiment for robotics Czech Technical University (Prague, Czechia) Embodied Cognition and Sensorimotor Contingency Theory for visual robotics.
2019 - 2022	Ph.D. Computer Vision for Robotics (President's Doctoral Scholar Award) The University of Manchester - Italian Institute of Technology (IIT) Bio-inspired visual attention models exploiting event-driven sensing and neuromorphic platforms for the robot iCub. Motion estimation, depth extraction, figure-ground segmentation, ego-motion segregation. - R&D & Project management Ph.D. Supervisors: Dr. Chiara Bartolozzi (IIT) & Prof. Angelo Cangelosi (UOM) <u>VIVA Ph.D. Examiners</u> : Prof. Giacomo Indiveri (ETH) & Prof. Piotr Dudek (UOM)
2018 - 2019	Software Engineer - Computer Vision for Robotics Italian Institute of Technology (Genoa, Italy) ◦ Software engineer, Middleware project combining Yarp Network, IBM Cloud services and Vodafone 5G for a robotic conversational healthcare application with IBM and Vodafone Italia. ◦ Software engineer, Computer Vision, stealth project with Sony Japan. Programming Languages and Tools: JavaScript, Matlab, C++, IBM Cloud services
2016 - 2017	Research Fellow, Computer Vision - Predictive Coding & Robotics King's College of London (London, UK) An egocentric neuromorphic network of the 3D space through supervised learning.

Technology Transfer - R&D & Project management

2025	CTU, Wandering around: A bioinspired approach to visual attention through object motion sensitivity Vision-Robotics Real-time selective attention through object motion sensitivity. Demo link
2024	CTU-Lille Uni., Event-driven nearshore and shoreline coastline detection on SpiNNaker neuromorphic hardware Vision-Robotics Event-based spiking neural network on the neuromorphic hardware SpiNN-3. Demo link
2023	ETH-IIT, Robotic sensory-motor paint brush Robotics Event-based trajectory painting on neuromorphic hardware. Demo link
2022	IIT, Event-driven Proto-object based saliency in 3D space Vision & Robotics Event-based depth extraction for visual attention. Demo link
2020	IIT, Proto-object based saliency for event-driven cameras Vision & Robotics Event-based visual attention. Demo link
2018, 2019	IIT-IBM, Robotic conversational application for health care Human-Robot Interaction Lead software engineer of the JointLab IIT-IBM, and Vodafone Italia. This project has been showcased at IROS 2018 & Milan Digital Week Event 2019. Demo link & Github

Projects' Demos 00 [Youtube link](#)

Editor & Communication Chair

I firmly believe that effective communication is about conveying a single, clear message with passion. I am the official Editor of **NeuroPAC** (in collaboration with The University of Maryland), Chair of **Open Neuromorphic** and co-organiser of **Telluride Neuromorphic Engineering Workshop**.

Women&Technologies Young Ambassador

Women&Technologies - Associazione Donne e Tecnologie. I advocate for women in STEAM at both national and international events.

Gender dimension & Diversity inclusion

I foster gender equality and diversity, encouraging open discussions and a safe environment ensuring **fairness, openness**, and addressing microaggressions. In my career, equality is a top priority.

Education

2019 - 2022	PhD in Computer Science (Neuromorphic algorithms) with honors	The University of Manchester
2015 - 2017	MSc Degree in Neuroengineering with honors	University of Genoa
2011 - 2015	Bachelor's Degree in Biomedical Engineering	University of Genoa

Competitive Grants & Awards

2025	Tecnovisionarie® 2025	International Award
	Female researchers and scientists under 40 leading the way in AI, new materials, space, energy and the environment, health, and biotechnology.	
2025	Nature Communications Editors' Highlights: "Event-Driven Figure-Ground Organisation model for the humanoid robot iCub"	Nature Communications
	Among the best 50 papers in Applied physics and mathematics	
2024	Marie Skłodowska-Curie Actions Award - Postdoctoral Fellowship	European Union
	EveNt DrivEn Active Vision for Object peRception (ENDEAVOR)	
2023	Best Oral Presentation	Women in Vision UK
	Winter Meeting	
2023	President's Doctoral Scholar Award	The University of Manchester
	Highest accolade bestowed upon postgraduate members. Academic excellence and leadership potential.	
2022	FameLab National Finalist	FameLab
	International Scientific Talent Show, 3 minutes scientific speech	
2022	NEUROTECH EU Fellowship	Neuromorphic Computing Technology community in Europe
	European Fellowship to attend Capocaccia Neuromorphic Workshop 2022	
2019	HiPEAC EU Collaboration Grant	SNN neuromorphic attention model
	European collaboration Grant, 3 months collaboration project finalised with the publication: "Event-driven bio-inspired attentive system for iCub humanoids robot on SpiNNaker" (DOI: 10.1088/2634-4386/ac6b50)	

Workshop & Conference organisation

2025	(ongoing) Organising committee	Neuromorphic Computing for Development and Learning Workshop (NCDL)
	International Conference on Development and Learning IEEE (ICDL)	
	Prague, Czech Republic	
2025	(ongoing) Organising committee	Workshop on Neuromorphic Perception for Real World Robotics (NeuRobots)
	IEEE/RSJ International Conference on Intelligent Robots and Systems IEEE (IROS)	
	Hangzhou, China	
2025	(ongoing) Media and Communication Chair, Organising committee	IEEE ICDL 2025
	International Conference on Development and Learning	
	Prague, Czech Republic	
2025,2024	(ongoing) Group Leader	Telluride Neuromorphic Engineering Workshop
	Telluride, Colorado, USA	
2024, 2023	Program committee, Organising committee	International Conference on Neuromorphic Systems (ICONS)
	Santa Fe, New Mexico, USA	
2023	Program committee, Organising committee	Living Machines (LM) Conference
	Tutorial 'Real-time motion estimation with SNNs', Genoa, Italy	

Seminars & Lectures

Other professional experiences

Professional Dancer & Dance Teacher 2009 - NOW

I have been studying dance since 1994. I am a professional dancer and teacher of Lindy Hop, Jazz steps and Swing social dance, Hip-Hop & House Dance. Winner of national and international competitions: Yo! Festival (Rome 2012), Au delà des préjugés (Lausanne 2015), ect). Associazione Italiana Maestri di Ballo (AIMB), professional dance teacher certificate. **Voluntary paramedic** 2014 - 2019 Public Assistance, national first aid certificate, 118 Certificate

2025, 2023	Event Based Revolution for Robotic Perception Prague, Czech Republic	Czech Technical University
2024	Bioinspired and neuromorphic algorithms for active vision Prague, Czech Republic	Charles University
2024	Saccadic revolution - Bioinspired active vision for robotics Sussex, United Kingdom	The University of Sussex
2021	The Neuromorphic iCub, Event-Driven Perception for Robotics Genoa, Italy	University of Genoa

Invited talks

2025	Neuromorphic active vision for embodied object perception INVICTA School - INESC TEC, Campus da Faculdade de Engenharia da Universidade do Porto Porto, Portugal	
2025	Awareness, adoption, and directions of neuromorphic technologies, STANCE Fraunhofer-Institut für Integrierte Schaltungen IIS Nuremberg, Germany	
2025	Embodied Active Vision for robotics Berlin, Germany	Embodied Agents in Contemporary Visual Art
2024	Rising Star Researcher Keynote - Neuromorphic algorithms to model visual attention ECCV 2024, Neuromorphic Vision Workshop Milan, Italy	
2024	Saccadic Revolution Sussex, UK	The University of Sussex
2024	Bioinspired and neuromorphic algorithms for robotics Prague, Czechia	IEEE Annual Meeting - Czechoslovakia
2024	Neuromorphic algorithms to model visual attention International School on Artificial Intelligence for Cognitive Technologies Naples, Italy	
2023	Bioinspired intelligent visual attention system for the humanoid robot iCub exploring event-driven sensing and neuromorphic hardware Cambridge	Women in Vision UK Winter Meeting
2023	Event-driven and neuromorphic visual attention algorithms for online robotic applications IEEE Instrumentation & Measurement Society Workshop Nancy, France	
2023	What's catching your eye? Online	Open Neuromorphic
2023	L'intelligenza dei robot ("The intelligence of the robots") Online	Women&Tech
2023	Bioinspiration, is it needed? Bioinspired and neuromorphic systems to model visual attention mechanisms Germany	Technische Universität Berlin
2023	The Event-Based Revolution XIV Symposium on Bioengineering, FEUP - Faculdade de Engenharia da Universidade do Porto Porto, Portugal	
2022	"Attenzione" (Attention) Perugia, Italy	FameLab Finalist
2022	What is digital technology? Milan, Italy	Elle Active!2022
2022	What attracts your attention? DTU, Copenhagen, Denmark	Technical University of Denmark

Active Collaborations

2019 - NOW

Prof. Ernst Niebur Johns Hopkins University - "Saliency-based bioinspired attention and active vision"
Dr. Chiara Bartolozzi Italian Institute of Technology - "Neuromorphic visual perception and embodiment"
Prof. Giacomo Indiveri ETH Zurich - "Neuromorphic algorithms and computing for vision"
Prof. Yulia Sandamirskaya Zurich University of Applied Sciences (ZHAW) - "Neuromorphic HRI"
Prof. Jan Antolik Charles University - "Modelling vision"
Dr. Dylan Muir Synsense - "Neuromorphic vision on Speck"
Dr. Alexander Hadjiivanov Netherlands eScience Center - "Understanding the retina layers"
Dr. Mazdak Fatahi & Prof. Pierre Boulet - Université de Lille - "SNNs on neuromorphic HW"
Prof. James Knight The University of Sussex - "Neuromorphic algorithms for navigation on GeNN"
Prof. Michael Furlong The University of Waterloo - "SNNs for active vision control"

Editorial & Reviewing Work

I am an **IOP Trusted Reviewer** for the journal Neuromorphic Computing and Engineering. I have also reviewed work for journals such as Neurocomputing (Springer Nature) and Frontiers in Neuroscience, as well as conferences including IEEE CVPR, IEEE IROS, IEEE/ACM ICONS, IEEE BIOCAS, and IEEE AICAS.

Academic Supervision

2019 - NOW

Tutoring and mentoring of several M.Sc. and PhD students. Skills learnt: routing, communication, active listening and leadership.

PhD, Riccardo Pignari "Object motion sensitivity", Università di Torino, PhD student, Collaboration with Prof. Gianvito Urgese

PhD, Caterina Caccavella "Neuromorphic vision for robotic applications", Zurich University of Applied Sciences (ZHAW), PhD student, Collaboration with Prof. Yulia Sandamirskaya

PhD, Akwasi Akwaboah "Event-driven saliency-based attention for active vision", Johns Hopkins University, PhD student, Collaboration with Prof. Ralph Etienne-Cummings and Ernst Niebur

PhD, Pouya Abdollahzadeh "Event-driven visuotactile schema for iCub, the humanoid robot", University of Genoa, PhD student, Co-supervision with Dr. Chiara Bartolozzi

MTs, Lioba Schürmann "Event-driven visual model for a robotic sensory-motor paint brush on neuromorphic hardware", Institute of Neuroinformatics (INI) ETH Zurich, MT student, Co-supervision with Prof. Giacomo Indiveri

MTs, Priyadarshini Kannan "Event-driven proto-object saliency on Loihi", Elite Masters of Neuroengineering at Technical University of Munich, MT student, Collaboration with Intel Labs

MTs, Arianna Gardella "Segregation of object and background motion in the retina", University of Genoa, MT student, Co-supervision with Dr. Chiara Bartolozzi

MTs, Simone Voto "Event-Driven Figure-Ground Organization", University of Genoa, MT student, Co-supervision with Dr. Chiara Bartolozzi

MTs, Luna Gava "Event-Driven Motion-In-Depth for Low Latency Control of the Humanoid Robot iCub", University of Genoa, MT student, Co-supervision with Dr. Chiara Bartolozzi

MTs, Ella Janotte "Biological Plausibility of the Spiking Elementary Motion Detector Model", Bielefeld University, MT student, Co-supervision with Prof. Elisabetta Chicca

MTs, Suman Ghosh "Event-driven bio-inspired online depth estimation for scene exploration on the iCub", European Master on Advanced Robotics, MT student, Co-supervision with Prof. Fabio Solari

Co-Founder & Co-Creator Brains&Machines Podcast

2023-NOW; A podcast about **neuromorphic engineering** and biologically-inspired technology, in collaboration with Prof. & Journalist Sunny Bains, **University College London** and Prof. Ralph Etienne Cummings, **Johns Hopkins University**. Sponsored by **EE Times | Electronic Engineering Times**.

Link: <https://brainsandmachines.net/>





Co-Founder & Co-Creator Sottosoglia Podcast

2023-NOW; A podcast about "The person behind the scientist", in collaboration with Dr. & Fabrizio Ottati, **NXP Semiconductors**. Sponsored by **Open Neuromorphic**.

Link: <https://www.youtube.com/@Sottosoglia/videos>

Attended Schools & Conferences (Selected)

- (2025) **Awareness, adoption, and directions of neuromorphic technologies, STANCE**, Fraunhofer-Institut für Integrierte Schaltungen IIS Nuremberg, Germany
- (2025) **Embodied Active Vision for robotics Embodied Agents in Contemporary Visual Art**, Berlin, Germany
- (2024) **European Conference on Computer Vision (ECCV)**, Milan, Italy
- (2023) **Living Machines**, International Conference in Biomimetic and Biohybrid systems, Genoa, Italy
- (2022) **NeuTouch**, International School on Technologies for Touch, Arenzano, Italy
- (2022, 2019) **CapoCaccia Neuromorphic Workshops**, CapoCaccia Workshops toward Neuromorphic Intelligence, Alghero, Italy
- (2022) **Neuro-Inspired Computational Elements (NICE)** Neuromorphic Workshop, Online
- (2021) **IEEE Women in Engineering**, University of Genoa, Italy
- (2020) **Neuromatch Academy**, NMA, Online
- (2019) **Machine Learning Crash Course**, University of Genoa, Italy
- (2019) **Summer School in Computational and Theoretical Models in Neuroscience** (ConTaMiNEURO), Venice, Italy

Publications List

Last Author - International peer-reviewed journals

- Event-driven nearshore and shoreline coastline detection on SpiNNaker neuromorphic hardware
Neuromorphic Computing and Engineering IOP science
DOI:10.1088/2634-4386/ad76d5

First Author / First Co-Author – International Peer-Reviewed Journals

- (Accepted – To Be Published) Wandering around: A bioinspired approach to visual attention through object motion sensitivity
Rising Stars 2025 Collection – Neuromorphic Computing and Engineering, IOP Science
- (Accepted – To Be Published) A neuromorphic electronic artist for robotic painting
Scientific Reports
DOI: 10.21203/rs.3.rs-4528779/v1
- Event-Driven Figure-Ground Organisation model for the humanoid robot iCub
Nature Communications
Featured as Editors' Highlights in Applied Physics and Mathematics
DOI: 10.21203/rs.3.rs-3897126/v1
- Event-driven bio-inspired attentive system for iCub humanoid robot on SpiNNaker
Neuromorphic Computing and Engineering, IOP Science
DOI: 10.1088/2634-4386/ac6b502022
- Event-Based Eccentric Motion Detection Exploiting Time Difference Encoding
Frontiers in Neuroscience
DOI: 10.3389/fnins.2020.00451
- Event-driven Proto-object based saliency in 3D space to attract a robot's attention
Scientific Reports
DOI: 10.1038/s41598-022-11723-6

Personal Statement

According to law 679/2016 of the Regulation of the European Parliament of 27th April 2016, I hereby express my consent to process and use the data provided in this CV.

Second Author – International Peer-Reviewed Conferences

- Proto-object based saliency for event-driven cameras
IEEE IROS
DOI: 10.1109/IROS40897.2019.8967943

First Author – Abstract

- Bioinspired intelligent visual attention system for the humanoid robot iCub exploring event-driven sensing and neuromorphic hardware
DOI: 10.1136/bmjoo-2024-WVUK.10